

Problem set 1

Econ 5030

1 Preferences and Utility

1. Assume that $\succsim$ is complete and transitive on X . Show the following properties must hold for $\sim$ and $\succ$.
 - (a) $x \succ x$ never holds,
 - (b) if $x \succ y$ and $y \succ z$, then $x \succ z$,
 - (c) $x \sim x$ for all x ,
 - (d) if $x \sim y$ and $y \sim z$, then $x \sim z$,
 - (e) if $x \sim y$, then $y \sim x$,
 - (f) if $x \succ y$ and $y \succsim z$, then $x \succ z$.
2. Take the set $X = \{a, b, c, d\}$. Suppose that $\succsim$ is complete and transitive on X .
 - (a) Prove that the set of optimal choices $C(X, \succsim) = \{x \in X : x \succsim y \text{ for all } y \in X\}$ is non-empty.
 - (b) Prove that $x \sim y$ for all $x, y \in C(X, \succsim)$.
3. Take the set $X = \{a, b, c, d\}$.
 - (a) Prove that if $\succsim$ is complete and transitive on X , then there exists a function $u : X \rightarrow \mathbb{R}$ that represents the preferences $\succsim$ on X .
 - (b) Explain why we need an additional assumption to get a utility representation if $X' = [0, 1]^2$. Give an example of preferences that are complete and transitive on X' that cannot be represented by a utility function.
4. Consider the utility-maximization problem on $X = \mathbb{R}_+^2$ with Cobb–Douglas utility

$$u(x_1, x_2) = x_1^{1/2} x_2^{1/2},$$

where $p = (p_1, p_2) \in \mathbb{R}_{++}^2$ and $w > 0$.

- (a) Show that an optimal bundle is interior and exhausts the budget. Use the first-order conditions to derive the Marshallian demand $D(p, w)$ and the indirect utility function $v(p, w)$.

- (b) Compute the demanded bundle and indirect utility when $(p_1, p_2, w) = (4, 1, 80)$.
- (c) Verify Walras' law and degree-zero homogeneity of demand at the values in part (b), using $\lambda = 2$.

5. Consider the utility-maximization problem on $X = \mathbb{R}_+^2$ with perfect-complements utility

$$u(x_1, x_2) = \min\{x_1, 2x_2\},$$

where $p = (p_1, p_2) \in \mathbb{R}_{++}^2$ and $w \geq 0$.

- (a) Derive the Marshallian demand $D(p, w)$ and the indirect utility function $v(p, w)$. Give an economic explanation for the proportions in the optimal bundle.
- (b) Compute the demanded bundle and indirect utility when $(p_1, p_2, w) = (3, 2, 80)$.
- (c) Show that the preferences represented by u are convex but not strictly convex. Explain why the demand correspondence is nevertheless a singleton (and hence convex-valued).